# Picking Winners in Factorland:
## A Machine Learning Approach to Predicting Factor Returns

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### KEY FINDINGS

- The article shows strong cross-sectional predictability of factor returns using machine learning techniques, focusing on past returns, risks, and spreads.
- Factors with the highest expected returns outperform the lowest by up to 1.39% per month, although this predictability gradually declines over time.
- Factor momentum is the main driver of cross-sectional variation in anomaly returns, capturing most of the predictability.

### ABSTRACT

The article examines the cross-sectional predictability of factor returns. Using machine learning techniques, the authors extract information from a comprehensive set of factor characteristics measuring their past returns, risks, and spreads. Applying this method to a repertoire of US stock market anomalies, they find robust predictability in their performance. The decile of factors with the highest expected return outperforms those with the worst outlook by up to 1.39% per month. The alphas are robust to many considerations but gradually decline over time. Predictability is mainly driven by factor momentum, which captures most of the cross-sectional variation in anomaly returns.

The rise of the anomaly literature has reshaped the investment landscape, bringing quantitative factor investing out of hedge fund trading desks and into the mainstream. With the proliferation of factor and smart beta exchange-traded funds (ETFs), factors are gradually replacing individual stocks as the building blocks of equity portfolios. But this fundamental shift raises new questions. How to navigate this anomaly zoo? How can we separate future winners from losers? Recent literature has just begun to identify signals for predicting factor performance. Examples include momentum (Ehsani and Linnainmaa 2022), seasonality (Keloharju, Linnainmaa, and Nyberg 2016), value spread (Zhang 2023; Anginer et al. 2024), and beta (Blitz 2024). Yet, a comprehensive framework for capturing cross-sectional variation in factor returns remains to be developed.

In this article, we offer a holistic approach to forecasting the cross section of factor returns. Our analyses leverage five decades of data on the 153 long–short anomaly portfolios in the US market from Jensen, Kelly, and Pedersen (2023). Using popular machine learning techniques, we extract information from 242 factor characteristics that reflect past returns, characteristic spreads, and risk measures. Rather than

arbitrarily selecting predictors, we let the data speak and flexibly form the forecasting function. Our goal is to assess the inherent predictability within the factor universe and its potential to pave the way for effective factor selection strategies. Finally, we seek to identify what drives the cross-sectional predictability of factor returns.

Our results can be summarized as follows. First, we demonstrate the substantial cross-sectional predictability of factor returns. Simple penalized regressions, such as least absolute shrinkage and selection operator (LASSO) or elastic net (ENET), generate out-of-sample predictive $R^2_{OOS}$ coefficients for monthly returns close to 0.6%. This magnitude of return predictability at the factor level compares favorably with the evidence at the US equity level (e.g., Gu, Kelly, and Xiu 2020) and exceeds the $R^2_{OOS}$ values obtained by much more complex models. Interestingly, in contrast to the stock universe, accounting for nonlinearities in interactions adds little value. Whether because of a dearth of data to effectively capture such nuanced patterns or their absence in the factor zoo, algorithms such as neural networks or regression trees fail to meaningfully improve predictability.

Second, we show that return predictability can be translated into successful factor selection strategies. The decile of factors with the highest returns outperforms the decile with the worst prospects by 0.27% to 1.39% per month, depending on the prediction algorithm. For the forecast combination model, which aggregates the predictions of all the individual models in our study, the monthly return on such a spread portfolio is 1.08%.

Notably, these cross-sectional patterns in anomaly returns are remarkably robust. They cannot be explained by common asset pricing factors, but they also outperform a naive benchmark that equally weights all anomalies in the sample. Moreover, the alphas are high and significant regardless of the specific implementation choices and hold for different break points used to construct portfolios at both the stock and factor levels. The predictability of returns, however, has decreased visibly over time. Specifically, although the returns and alphas of the machine learning strategies remain sizable and significant over the past two decades, their magnitude has declined by about half compared to earlier years.

Third, we identify the key sources of factor return predictability. A variable importance analysis reveals that the cross-sectional variation in factor returns is driven by a handful of key characteristics, with short-term factor momentum playing a fundamental role. For example, the one-month factor momentum advocated by Avramov et al. (2017) and Gupta and Kelly (2019), among others, frequently takes prime positions in the importance ranking. The simple momentum spread, that is, the difference between the historical returns of stocks within long and short legs, also ranks highly. Other characteristics, such as characteristic (or value) spreads or risk measures, play a minor role.

To better understand the role of factor momentum in shaping the predictability of returns, we perform spanning tests of machine learning strategies against factor momentum portfolios. The regressions unveil the essential role of factor momentum. The machine learning strategies exhibit substantial exposure to factor momentum portfolios, which explains almost all of their abnormal returns, rendering them insignificant. Specifically, once we control for factor momentum, no long–short machine learning portfolio generates significant alpha. In sum, the factor momentum effect emerges as the key driver of the cross-sectional variation in factor performance, capturing most of the existing return predictability.

Our article relates to two main strands of academic literature. First, we extend the recent debate on the return predictability of factor returns. Previous studies have largely focused on documenting individual characteristics that capture the variation in factor returns. These include, for example, factor momentum (Avramov et al. 2017; Ehsani and Linnainmaa 2022; Fan et al. 2022; Arnott, Kalesnik, and Linnainmaa 2023),

unconditional mean (Leippold and Yang 2021), seasonality (Keloharju, Linnainmaa, and Nyberg 2016), value spread (Haddad, Kozak, and Santosh 2020; Anginer et al. 2024), issuer-repurchaser spread (Greenwood and Hanson 2012), volatility (Moreira and Muir 2017; Cederburg et al. 2020), or beta (Blitz 2024). Few studies have attempted to integrate information from multiple signals—primarily in the context of factor timing and time-series return variation (e.g., Galakis, Vrontos, and Vrontos 2021; Kagkadis et al. 2024; Neuhierl et al. 2023). To our knowledge, this article is the first to examine the aggregate cross-sectional predictability of factor returns.

Second, our article adds to the rapidly growing body of research on machine learning applications to the cross section of asset returns. Previous evidence has covered diverse asset classes, including individual stocks from the US and international markets (Aw, Jiang, and Jiang 2019, 2022; Simonian et al. 2019; Gu, Kelly, and Xiu 2020, 2021; Leippold, Wang, and Zhou 2022; Avramov et al. 2017; Azevedo, Kaiser, and Müller 2023; Blitz et al. 2023; Cakici et al. 2023a; Fieberg et al. 2023; Hanauer and Kalsbach 2023), government bonds (Bianchi, Büchner, and Tamoni 2021), corporate bonds (Bali et al. 2020), commodities (Rad et al. 2023), or options (Bali et al. 2023). Applications at the meta (portfolio) level are more scarce and mainly limited to industries (Rapach et al. 2019) and country indexes (Cakici and Zaremba 2023). As far as we are concerned, our article is the first attempt to use machine learning techniques to extract information on the cross section of factor returns.

The remainder of this article proceeds as follows. The second section presents the data and methods. The third section discusses the main empirical findings. The fourth section zooms onto the role of factor momentum. Finally, the fourth section concludes the study.

## DATA AND METHODS

In this section, we introduce the sample of stocks and factor portfolios used in the study. Next, we discuss the forecasting models used to predict factor returns.

### Data Sources and Sample Preparation

Our sample of US factor returns builds on the 153 stock characteristics from Jensen, Kelly, and Pedersen (2023). This comprehensive set encompasses the most prominent anomalies documented by finance literature (see Appendix for the list of covered characteristics). To avoid any arbitrariness, we reproduce all the return predicting signals and calculate them by closely following the procedures in Jensen, Kelly, and Pedersen (2023) and using their original replication code available from github.com/bkelly-lab/ReplicationCrisis.

The same code is also used to gather all underlying data, with the accounting data coming from Compustat and market data from CRSP. We prepare the data using typical methods from asset pricing literature and closely reproduce the filters from Jensen, Kelly, and Pedersen (2023). In particular, we restrict the sample to common stocks, which Compustat identifies as primary securities of the underlying companies. The study period runs from January 1972 to December 2021, with the start date being determined by the availability of data necessary to implement the machine learning models.

### Factor Portfolios

We construct a comprehensive set of factor characteristics that have been identified in the asset pricing literature to predict factor returns. Importantly, rather than

## EXHIBIT 1

### Factor Characteristics

| Characteristic | Symbol | Definition | Variants |
|---|---|---|---|
| Past Monthly Return | ret_m_x | The monthly return lagged by *x* months, where *x* can take values from 1 to 60. For example, ret_m_3 indicates the monthly factor return for month *t* – 3. | 60 |
| Past Annual Return | ret_y_x | The annual return lagged by *x* years, where *x* can take values from 6 to 20. For example, ret_y_7 indicates the annual return for the year *t* – 7. | 15 |
| Daily Volatility | vol_d_xm | The standard deviation of daily returns observed over the past *x* months, where *x* can take values of 1, 3, 6, or 12 months. For example, vol_d_3m indicates the return standard deviation estimated using three months of daily data. | 4 |
| Monthly Volatility | vol_m_xy | The standard deviation of monthly returns observed over the past *x* years, where *x* can take values of 2, 3, 4, or 5 years. For example, vol_m_3 indicates the return standard deviation estimated using three years of monthly data. | 4 |
| Characteristic Spread | spread_x | Characteristic spread for factor *x* calculated as a difference between the value-weighted average characteristic in the long and short leg. spread_ be_me indicates the book-to-market ratio spread. | 153 |
| Market Beta | beta_xm | The portfolio's beta, estimated as the slope coefficient from a regression of the portfolio's returns on the excess returns of the market portfolio, using data from the past *x* months, where *x* can be 12, 24, 36, 48, 60 months, or all available historical data. For example, beta_24m indicates the beta based on 24-month rolling estimation period. | 6 |
| Total Number of Factor Characteristics | | | 242 |

NOTES: This exhibit lists the characteristics used to predict factor returns. The last column shows the total number of variable variants given the possible values of *x* in the definition.

replicating precise feature specifications from the prior literature, we compute broader sets of features based on alternative data frequencies and estimation methods. This approach aims to minimize arbitrariness in feature selection by letting the data speak. Overall, the features considered can be grouped into three broad categories: past return, characteristic spread, and risk. Exhibit 1 illustrates the specific definitions of the variables under consideration, while a broader context is provided in the following.

**Past returns.** A number of studies relate factor performance to their past returns, although the specific length, lag, and sign of the correlation with future returns may vary. For example, several papers document a strong factor momentum effect (Avramov et al. 2017; Gupta and Kelly 2019; Leippold and Yang 2021; Ehsani and Linnainmaa 2022; Fan et al. 2022; Arnott, Kalesnik, and Linnainmaa 2023; Yan and Yu 2023; Ma, Liao, and Jiang 2024), claiming that past factor returns predict their future performance. The exact length of the measurement period can vary because Ehsani and Linnainmaa (2022) rely on a 12-month estimation period, while Avramov et al. (2017) and Gupta and Kelly (2019) emphasize the special role of the last month. In addition, Leippold and Yang (2021) emphasize the importance of the long-run (unconditional mean), and Keloharju, Linnainmaa, and Nyberg (2016) and Blitz (2024) document the cross-sectional seasonality effect in anomaly returns, showing that average returns of the same calendar month help predict future factor performance.

All of the signals previously mentioned rely on some form of weighting of past returns. To avoid any arbitrariness in the selection, we include simply all individual monthly returns over the last five years (*t* – 60 to *t* – 1), as well as annual returns for the preceding 15 years. We assume that the machine learning model will flexibly and automatically select and weigh the most important past returns, uncovering the true structure of return dependence.

**Characteristic spread.** A large body of literature has shown that expected returns on the spread in valuation ratios tend to comove with expected returns on the

value-minus-growth strategy (Asness et al. 2000; Cohen, Polk, and Vuolteenaho 2003; Blitz and Hanauer 2020; Haddad, Kozak, and Santosh 2020). Recent studies, however, extended this effect outside the stock market (Baba Yara, Boons, and Tamoni 2021) and to other anomalies, that expensive (cheap) factors in terms of their value spread tend to produce relatively lower (higher) returns (Anginer et al. 2024), though sometimes with mixed results (Asness 2016; Asness et al. 2017). From a broader perspective, the value spread is just one specific case of characteristic spread, which could be extended to other differences in other stock characteristics between the long and short anomaly leg. For example, Zhang (2023) documents the role of the spread on long-term stock returns, Greenwood and Hanson (2012) find that the magnitude of share issuance matters, and Huang (2022) finds that the return spread between winners and losers negatively predicts momentum factor returns. Finally, Huang et al. (2010) argue that the spread in the (sorting) characteristic between the top and the bottom quantiles intuitively reflects future return dispersion.

Based on this evidence, we consider all possible characteristic spreads for each factor. In other words, rather than limiting our consideration to a few selected variables, we calculated the differences between the average values of each of the 153 stock characteristics for the long and short anomaly portfolios. This gives us 153 characteristic spreads for each factor.

**Risk.** Several papers document that certain proxies for risk may affect the return predictability of anomalies, although this is not without controversy. For example, Daniel and Moskowitz (2016) show that volatility can predict the downside of momentum. Factor volatility as a timing signal is controversial. DeMiguel, Martin-Utrera, and Uppal (2021), building on the framework of Brandt, Santa-Clara, and Valkanov (2009), show the outperformance of a conditional mean–variance multifactor portfolio with weights adjusted to market volatility. Moreira and Muir (2017) find that past factor volatility assists in the selection of time-varying exposures to factors, particularly the market. Cederburg et al. (2020) and Barroso and Detzel (2021) suggest that the performance benefits of volatility management diminish with realistic assumptions, such as trading costs. Reschenhofer and Zechner (2022) suggest using past factor return volatilities for factor timing. Although all of the preceding measures focus on total volatility, Blitz (2024) narrows the risk measure to its systematic component. As a result, he finds evidence of the counterpart of the low-beta anomaly at the factor level.

As for the other measures, we use several different risk proxies—also taking into account different estimation periods and frequencies—to avoid any arbitrariness in the feature selection. Specifically, we use i) daily volatility derived from the past 1, 3, 6, and 12 months; ii) monthly volatility based on the past 2, 3, 4, or 5 years of data; and iii) market beta calculated using various windows ranging from 12 months to all historical data.[1]

All factor portfolios are implemented in a consistent manner: Each month, we sort all stocks on a given characteristic and group them into deciles. We then build a long–short strategy that buys the decile of stocks with the highest expected returns and shorts those with the worst prospects. The portfolios are value weighted and rebalanced monthly.

## Prediction Models

We use machine learning techniques to extract the aggregate information contained in factor characteristics. These models allow us to process a large number

[1] In unreported analyses, we consider a number of adjustments for all 242 factor characteristics. Specifically, we experiment with standardizing them by their volatility and de-meaning them by their historical average. None of these operations, however, significantly improve their predictive ability.

of features and automatically identify those most relevant to future returns. They also effectively address overfitting concerns that can arise when dealing with a large number of potentially correlated predictors. Finally, sophisticated models can adeptly capture nonlinearities and interactions in the data that traditional methods may miss.

We employ a general additive forecasting model to capture the correlation between factor returns and their respective characteristics:

$$R_{i,t+1} = E_t(R_{i,t+1}) + \varepsilon_{i,t+1} \tag{1}$$

where $r_{R,t+1}$ is the return on factor $i = 1, \ldots, N_T$ in the month $t = 1, \ldots, T$. We calculate the expected returns, denoted as $E_t(r_{i,t+1})$, as a constant function of factor characteristics:

$$E_t(R_{i,t+1}) = g(z_{i,t}) \tag{2}$$

with $z_{i,t}$ denoting the vector of factor characteristics that includes the 242 variables described in Factor Portfolios section. The function $g(z_{i,t})$ estimates the returns based solely on the information available for factor $i$ at month $t$. This means that it ignores the characteristics of other factors or those from previous periods. In addition, its specific form is not explicitly defined. The approximated functions vary among machine learning models and can be either parametric or nonparametric, linear or nonlinear. We train all models to forecast future returns by minimizing the out-of-sample mean squared forecast error, and our overall goal is to identify a prediction model with superior forecasting accuracy from a pool of potential candidates.

Extensive machine-learning literature offers a wide range of prediction models. To mitigate potential selection bias linked to individual algorithm choices, we draw on several classical models commonly employed in prominent asset pricing studies, such as Gu, Kelly, and Xiu (2020); Leippold, Wang, and Zhou (2022); and Bali et al. (2020, 2023).

**Ordinary least squares (OLS).** OLS regression involves building a predictive model using all available features as inputs. This simple model does not require regularization, hyperparameter tuning, or validation. It is, however, prone to overfitting, especially in high-dimensional settings (Gu, Kelly, and Xiu 2020).

**Partial least squares (PLS).** PLS serves as an effective dimension reduction technique that accounts for the relationship between covariates and security returns. The three-pass PLS regression proposed by Kelly and Pruitt (2013, 2015) emphasizes characteristics that are highly correlated with stock payoffs and aggregates predictors into composite factors to maximize the correlation with future returns. The tuning parameter in this regression is the number of components.

**Penalized linear regression (LASSO, ENET).** Penalized linear regression combats overfitting by introducing a penalty term on the slope coefficients. We use two popular regularization approaches, LASSO and ENET introduced by Zou and Hastie (2005). LASSO penalizes based on the absolute values of the coefficients, while ENET combines LASSO and ridge regression, focusing on the squared coefficients. In our case, both components have the same weight. ENET is more effective than LASSO in dealing with covariate correlations (Zou and Hastie 2005; Diebold and Shin 2019). The penalty term serves as a tuning parameter in both models.

**Tree models (RF, GBRT).** Tree models such as random forests (RF) and gradient boosted regression trees (GBRT) provide flexibility in capturing interactions and nonlinearities in returns. They classify observations into subcategories or *leaves*. Strong regularization mitigates overfitting. RF uses bootstrap aggregation or *bagging* to average multiple trees, while GBRT fits subsequent trees based on residuals from previous trees. We use the least squares boosting method (Breiman 2001; Hastie, Tibshirani, and Friedman 2008) for GBRT, typically using between 100 and 200 learning cycles.

**Feed forward neural network (FFNN).** An FFNN consists of an *input layer* representing factor characteristics, variable *hidden layers* with activation functions, and an *output layer* that translates the results into return predictions. The neural network allows for modeling nonlinearities and interactions, with flexibility increasing with the number of layers. Following Bali et al. (2020), we use a single hidden layer FFNN with eight neurons and rectified linear units as the activation function and optimize the model using the Adam algorithm of Kingma and Ba (2014).

**Forecast combination (COMB).** The practice of combining multiple forecasts is based on well-established statistical principles (Bates and Granger 1969; Clemen 1989; Timmermann 2006). Forecast combination serves to reduce model variance, which ultimately leads to a reduction in forecast error. Following the approach outlined in Bali et al. (2020), we generate the COMB forecast by computing a simple equally weighted average of the predictions from seven different models mentioned earlier: OLS, PLS, LASSO, ENET, RF, GBRT, and FFNN.

For each model, we estimate parameters, tune hyperparameters, and evaluate predictive performance using standard machine learning methods. We apply a temporal sample split into an increasing training period, a fixed validation period, and an out-of-sample test period. The shortest in-sample period is 15 years, including 10 and 5 years for the training and validation windows, respectively, with the former gradually increasing. The training sample is used to estimate model parameters with prespecified hyperparameters, and the validation sample fine tunes the hyperparameters to minimize the objective loss function. We then test the accuracy of the model on a holdout dataset covering the subsequent 12 months to ensure an unbiased assessment on unseen data for out-of-sample evaluation.

Given the study period of January 1972 to December 2021 (600 months), the first training period is January 1972 to December 1981, the first validation period is January 1982 to December 1986, and the subsequent test period covers data from January to December 1987. Because of the computational intensity of machine learning models, we follow the approach of Gu, Kelly, and Xiu (2020) and Leippold, Wang, and Zhou (2022) and reestimate annually until the end of the sample. For each reestimation, we extend the training window by one year while keeping the validation and test periods fixed. The entire test period spans from January 1987 to December 2021, or 420 months.

## BASELINE EMPIRICAL FINDINGS

In this section, we summarize the essential empirical findings on the cross-sectional predictability of factor returns. We start with the assessment of prediction performance and continue with portfolio sorts. Lastly, we supplement our analysis with several additional robustness checks.

### Prediction Performance

To begin with, we assess the prediction performance with the out-of-sample $R^2$ coefficient, a popular measure frequently employed in asset pricing literature:

$$R^2_{OOS} = 1 - \frac{\sum_{(i,t)\in T_3} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{(i,t)\in T_3} r_{i,t+1}^2} \tag{3}$$

Here, $\hat{r}_{i,t+1}$ and $r_{i,t+1}$ represent the predicted and realized monthly returns on factor $i$ in month $t+1$, and $T_3$ denotes the testing sample. The calculation of $R^2_{OOS}$ uses the entire

## EXHIBIT 2

### Predictive Performance for the Cross Section of Factor Returns

| | OLS | PLS | LASSO | ENET | RF | GBRT | FFNN | COMB |
|---|---|---|---|---|---|---|---|---|
| $R^2_{OOS}$ | –4.025 | –1.034 | 0.601 | 0.593 | 0.526 | 0.616 | 0.087 | 0.479 |
| $\overline{\rho}_P$ | 0.030 | 0.052 | 0.070 | 0.070 | 0.047 | 0.067 | 0.045 | 0.060 |
| $\overline{\rho}_S$ | 0.036 | 0.058 | 0.079 | 0.079 | 0.065 | 0.085 | 0.045 | 0.070 |

**NOTES:** This exhibit reports the prediction performance measures for machine learning models applied to the cross section of factor returns: the out-of-sample $R^2$ coefficients ($R^2_{OOS}$) and average monthly Pearson ($\overline{\rho}_P$) and Spearman ($\overline{\rho}_S$) correlation coefficients between the predicted and realized returns. $R^2_{OOS}$ is expressed in percentages, and other values are decimals. The considered models include OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

sample of monthly returns and forecast over all testing periods, pooling observations across stocks and time. Higher values of $R^2_{OOS}$ indicate that the model's forecasts effectively capture the variation in factor returns.

Although $R^2_{OOS}$ is widely used to gauge forecasting accuracy, its interpretability or relevance may be limited in certain contexts. For example, it may fail to capture the model's ability to rank factors according to their actual ex post realized returns—essentially, its capacity to distinguish between winners and losers. The $R^2_{OOS}$ may obscure this link between predicted and realized returns because the cross-sectional correlation is overshadowed by return variances (Coqueret 2022). Consequently, investors could still achieve significant economic gains even when $R^2_{OOS}$ is negative (Kelly, Malamud, and Zhou 2021).

To address this limitation, we supplement $R^2_{OOS}$ with straightforward cross-sectional correlation coefficients. Concretely, we calculate time-series averages of monthly Pearson product-moment ($\overline{\rho}_P$) and Spearman rank-based ($\overline{\rho}_S$) correlation coefficients between predicted and realized returns. These additional metrics offer an intuitive overview of the relationship between model predictions and actual payoffs.

Exhibit 2 summarizes the prediction performance of different machine learning techniques. Overall, the models capture certain degrees of predictability, but the differences between them are substantial. Observe, for example, the first row focusing on $R^2_{OOS}$ measures. The values for OLS are strongly negative. OLS is well known for its tendency to overfit (see, e.g., Gu, Kelly, and Xiu 2020), which manifests itself also in this case. Being unable to cope properly with the multitude of factor characteristics, it yields the $R^2_{OOS}$ value of –4.025%. Notably, the application of PLS, a popular dimension reduction technique, leads to only a partial improvement: The $R^2_{OOS}$ remains negative, amounting to –1.034%.

The performance of LASSO and ENET indicates the ability to identify the relevant features among the characteristic zoo, which proves critical in factor return predictions. These two methods yield $R^2_{OOS}$ values of 0.601% and 0.593%, respectively. The difference between the simple and penalized regressions is remarkable.

Finally, the last four columns of Exhibit 2 show the results for more complex methods. Their common feature is the ability to account for nonlinearities and interactions between variables. These features, however, are not particularly helpful in the context of factor returns. Although the $R^2_{OOS}$ is still positive, they are mostly on par with penalized regressions. RF, GBRT, and COMB deliver qualitatively similar predictive performance, while the results for FFNN are somewhat worse. These disappointing numbers for these models, which typically overperform in stock market studies (e.g., Gu, Kelly, and Xiu 2020; Kelly, Malamud, and Zhou 2021), may signal two things. Either the nonlinearities and interactions are less important in the cross

section of factor returns than in individual stocks, or the panel of factor data is too small to capture such nuances in their performance. The relatively poor efficiency of FFNN, which is particularly computationally intensive and requires the estimation of many parameters, may lean toward the second hypothesis.

The bottom two rows of Exhibit 2 focus on the average cross-sectional correlation coefficient. Although the recorded values are not high, they confirm that the machine learning forecasts allow, to some extent, to sort the factors according to their realized (ex post) returns. This suggests that the prediction could potentially be used to effectively select the best-performing factors and avoid the market losers. Finally, the patterns in correlations across algorithms are generally consistent with those seen for the $R^2_{OOS}$ values: Correlations are particularly high for penalized regressions, and more complex methods do not add much value.

Finally, how much return predictability can be captured in factor returns relative to individual stocks? Overall, the observations compare favorably with previous evidence at the security level in the US market. For example, the best model in the seminal study by Gu, Kelly, and Xiu (2020)—a multilayer neural network—yields an $R^2_{OOS}$ of 0.40%. This is lower than the value of the simple penalized regressions in Exhibit 2. Notably, the study period in Gu, Kelly, and Xiu (2020) goes back to the 1950s, when markets were less efficient, providing more fertile ground for return predictability. Yet, the $R^2_{OOS}$ for the cross section of factor returns turns out to be higher. This suggests significant potential for factor-picking strategies, which we explore in the next section.

## Portfolio Sorts

To further illustrate the cross-sectional patterns of factor returns, we group them into decile portfolios. Specifically, each month we rank all factors by their expected returns, as determined by the machine learning models, and sort them into 10 buckets. We then create spread strategies, which involve going long the factors with the highest expected returns and shorting those with the worst outlook. Their returns can serve as an acid test of the monotonic patterns in the cross section of factor returns.

To evaluate the strategy's performance, we use two factor models. First, we want to test whether our active factor rotation strategies outperform a portfolio with static exposure to popular factors. Therefore, we apply Fama and French's (2018) six-factor model, which accounts for market, size, value, profitability, investment, and momentum effects:

$$R_t = \alpha_{F6} + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{WML} WML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \varepsilon_t \quad (4)$$

where $R_t$ is the month $t$ return on the test factor rotation strategy. $\beta_{MKT}$, $\beta_{SMB}$, $\beta_{HML}$, $\beta_{WML}$, $\beta_{RMW}$, and $\beta_{CMA}$ are estimated exposures to popular risk factors: market risk factor (*MKT*), small minus big (*SMB*), high minus low (*HML*), winners minus losers (*WML*), robust minus weak (*RMW*), and conservative minus aggressive (*CMA*), all computed as in Fama and French (2018). $\alpha_{F6}$ denotes the alpha, that is, the monthly abnormal return, and $\varepsilon_t$ is the error term.

Second, we juxtapose the machine learning strategies with an equal-weighted benchmark, which assumes allocating an equal amount of capital to each factor strategy in our sample. This helps to understand whether dynamic factor selection adds value relative to a naïve investment in all available factors:

$$R_t = \alpha_B + \beta_B B_t + \varepsilon_t \quad (5)$$

where $\beta_B$ denotes the monthly return on the equal-weighted benchmark, and $\alpha_B$ is the corresponding alpha.

## EXHIBIT 3

### Returns on Machine Learning Factors Portfolios

| | OLS | PLS | LASSO | ENET | RF | GBRT | FFNN | COMB |
|---|---|---|---|---|---|---|---|---|
| Low | 0.02 | –0.48 | –0.57 | –0.56 | –0.59 | –0.58 | –0.19 | –0.42 |
| 2 | –0.08 | –0.24 | –0.31 | –0.31 | –0.23 | –0.38 | –0.13 | –0.25 |
| 3 | 0.04 | –0.20 | –0.22 | –0.22 | –0.14 | –0.25 | –0.14 | –0.23 |
| 4 | 0.13 | –0.10 | –0.18 | –0.18 | 0.03 | –0.10 | 0.00 | –0.03 |
| 5 | 0.01 | 0.08 | –0.04 | –0.04 | 0.03 | –0.01 | –0.08 | 0.01 |
| 6 | 0.04 | 0.15 | 0.16 | 0.17 | 0.08 | 0.03 | 0.19 | 0.10 |
| 7 | 0.00 | 0.22 | 0.25 | 0.25 | 0.10 | 0.37 | 0.07 | 0.14 |
| 8 | 0.23 | 0.34 | 0.35 | 0.34 | 0.35 | 0.26 | 0.20 | 0.28 |
| 9 | 0.17 | 0.44 | 0.54 | 0.54 | 0.46 | 0.58 | 0.30 | 0.52 |
| High | 0.29 | 0.58 | 0.77 | 0.77 | 0.69 | 0.81 | 0.59 | 0.66 |
| High-Low | 0.27 | 1.06 | 1.34 | 1.33 | 1.29 | 1.39 | 0.78 | 1.08 |
| | *(0.74)* | *(3.29)* | *(3.87)* | *(3.82)* | *(3.79)* | *(4.31)* | *(2.34)* | *(3.51)* |
| $\alpha_{F6}$ | 0.32 | 1.11 | 1.30 | 1.29 | 1.19 | 1.34 | 0.63 | 1.04 |
| | *(0.91)* | *(3.76)* | *(4.37)* | *(4.35)* | *(3.96)* | *(4.31)* | *(2.82)* | *(3.37)* |
| $\alpha_{B}$ | 0.17 | 1.27 | 1.46 | 1.45 | 1.51 | 1.56 | 0.98 | 1.20 |
| | *(0.44)* | *(3.69)* | *(3.96)* | *(3.93)* | *(3.97)* | *(4.18)* | *(3.17)* | *(3.29)* |

**NOTES:** This exhibit reports monthly returns on decile portfolios of factor strategies formed on the predictions of machine learning models. The considered models include OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). Low (high) indicates the decile of factors with the lowest (highest) predicted return. High-low is the spread portfolio that assumes a long (short) position in the high (low) deciles. $\alpha_{F6}$ and $\alpha_{F6B}$ are alphas from the six-factor model of Fama and French (2018) and the factor benchmark model, respectively. All the returns and alphas are expressed in percentages. The numbers in parentheses are Newey and West's (1987) adjusted *t*-statistics. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

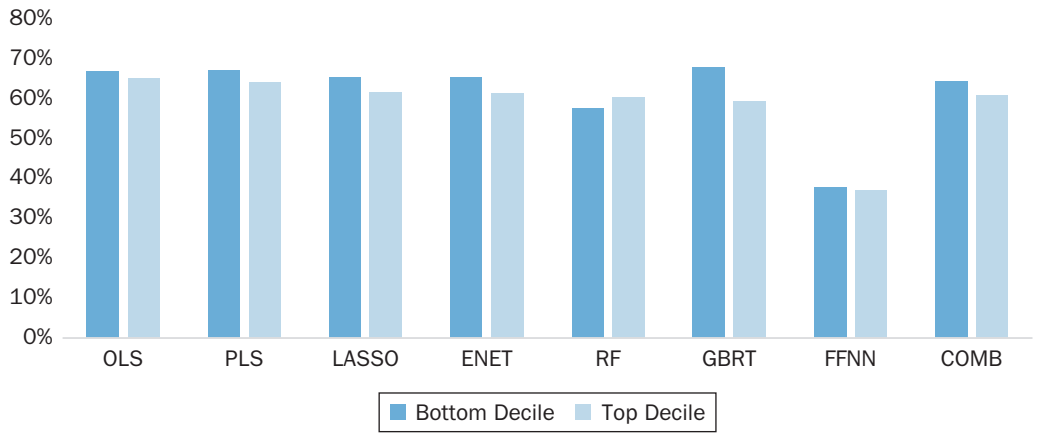
Exhibit 3 shows the performance of decile portfolios on machine learning predictions. Almost all models produce clear cross-sectional patterns in portfolio returns. The only exception is OLS, which also performed poorly in our previous tests. For all other models, the average return differences between the top and bottom deciles range from 0.76% (FFNN) to 1.39% (GBRT), all of which are statistically significant. In other words, machine learning models capture a significant amount of return predictability, allowing them to pick winners and avoid losers among factor strategies.

Among the different strategies, the penalized regression again takes the lead, mirroring the results in Exhibit 2. LASSO and ENET, with average monthly returns on spread portfolios of 1.33% and 1.29%, respectively, are beaten only by GBRT, and only by 5 basis points. Again, this highlights either the limited role of nonlinearities and interactions in the factor universe or the inability of standard machine learning models to capture them. Whichever is true, simplicity prevails and the complex models add little value.

The bottom rows of Exhibit 3 focus on the alphas of the factor models. The six-factor model clearly cannot explain the abnormal returns of the spread portfolios. Monthly alphas from algorithms other than OLS range from 0.63% (FFNN) to 1.34% (GBRT). The standard common factors fail to capture the cross-sectional patterns in factor returns, and the level of alphas is qualitatively consistent with the mean returns. Importantly, the benchmark model also fails to explain the abnormal returns of the spread portfolios. $\alpha_{B}$ values are remarkably high and significant for all models except OLS. This suggests that dynamic factor allocation strategies add value beyond simple static multifactor allocation. As a result, they deliver alphas even above a highly diversified benchmark that equally weights 153 factor strategies.

## EXHIBIT 4

### Portfolio Turnover of the Machine Learning Factor Strategies




**NOTES:** This exhibit reports average monthly turnover ratios for the decile portfolios of factor strategies formed on the predictions of machine learning models. The considered models include OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). Bottom (top) deciles are the portfolios of factors with the lowest (highest) predicted return. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

Harvey, Liu, and Zhu (2016) recommend more stringent significance hurdles to account for the increasing number of studies identifying patterns in the cross section of stock returns. To mitigate the risk of Type II errors, they suggest using a *t*-statistic threshold of 3. Our results in Exhibit 3 predominantly meet this criterion, with most alphas—excluding OLS—showing *t*-values exceeding 3 and, in some cases, even surpassing 4. This indicates that the abnormal returns we report are unlikely to be mere statistical artifacts.[2]

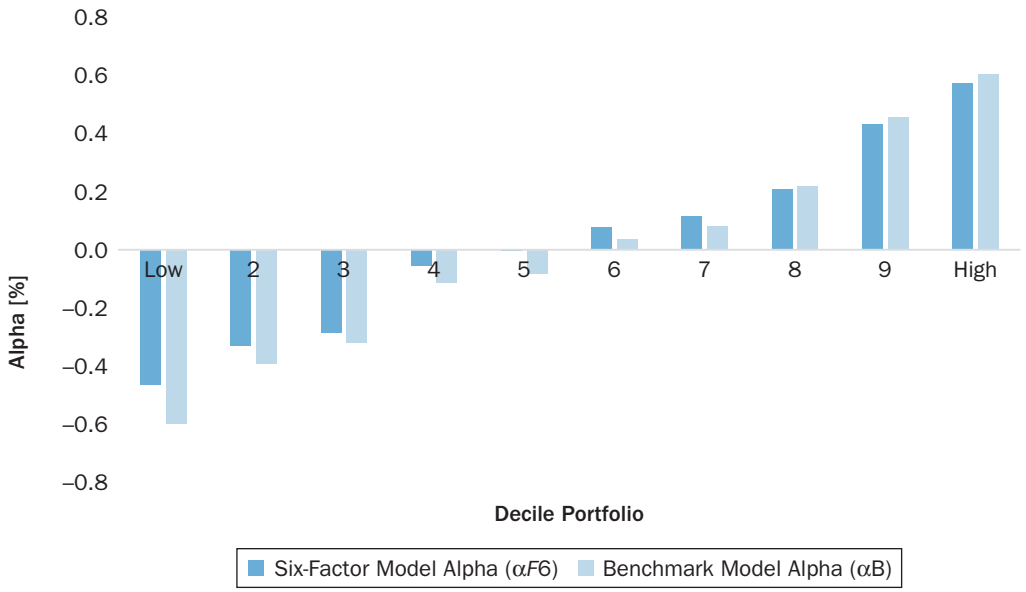
The advantage of machine learning strategies over the static benchmark lies in their ability to dynamically select factors with the best and worst outlook. How active is this rotation? To illustrate this, Exhibit 4 plots the average monthly cross-factor turnover for the top and decile benchmarks. Specifically, each month we calculate what proportion of factors is replaced by others and then compute a time-series average. Clearly, the machine learning strategies require substantial portfolio rotation. Specifically, each month, an investor needs to replace 37% to 66% of all factors, depending on the strategy.

Finally, we look at the magnitude of the abnormal returns across the decile sorts. Although Exhibit 2, for brevity, reports only the alphas of the spread portfolios, Exhibit 5 shows the abnormal returns for all decile portfolios. For conciseness, we limit our consideration to the COMB model, which aggregates the individual approaches. The plot shows clear monotonicity across all deciles for both the six-factor and benchmark models. In other words, the abnormal returns and return predictability do not come from any particular segment of the factor universe, such as the extreme losers or winners. The expected payoffs increase gradually and systematically as the machine learning forecasts improve.

[2] Jensen, Kelly, and Pedersen (2023) argue that the multiple testing framework recommended by Harvey, Liu, and Zhu (2016) may be overly conservative. They posit that the factor universe should not be viewed as a collection of hundreds of distinct factors but, rather, as clusters of a smaller number of correlated themes. As a result, they may not necessitate equally strict *t*-value thresholds across the board.

## EXHIBIT 5

### Alphas on the Combination Model Factor Portfolios




**NOTES:** This exhibit reports the monthly alphas on the decile portfolios of factors sorted on the forecast combination model (COMB), which aggregates seven individual machine learning models: OLS, PLS, LASSO, ENET, random forests (RF), GBRT, and FFNN. Low (high) indicates the decile of factors with the lowest (highest) predicted return. The alphas are obtained from the six-factor model of Fama and French (2018, $\alpha_{F6}$) and the factor benchmark model ($\alpha_B$). The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

### Further Robustness Checks

Having established the basic properties of factor return predictability, we turn to additional robustness checks. Specifically, we look at the performance of machine learning strategies over time and consider alternative portfolio construction break points.

**Performance through time.** A large body of financial literature points to the attenuation of market anomalies over time—particularly in the United States (Schwert 2003; Chordia, Roll, and Subrahmanyam 2011; McLean and Pontiff 2016; Zaremba, Umutlu, and Maydybura 2020; Cakici et al. 2023b). Regardless of the reasons—whether it is investor learning or improvements in market efficiency and liquidity—return predictability has declined.[3] But is the same pattern evident at the meta-level in the universe of factor portfolios?

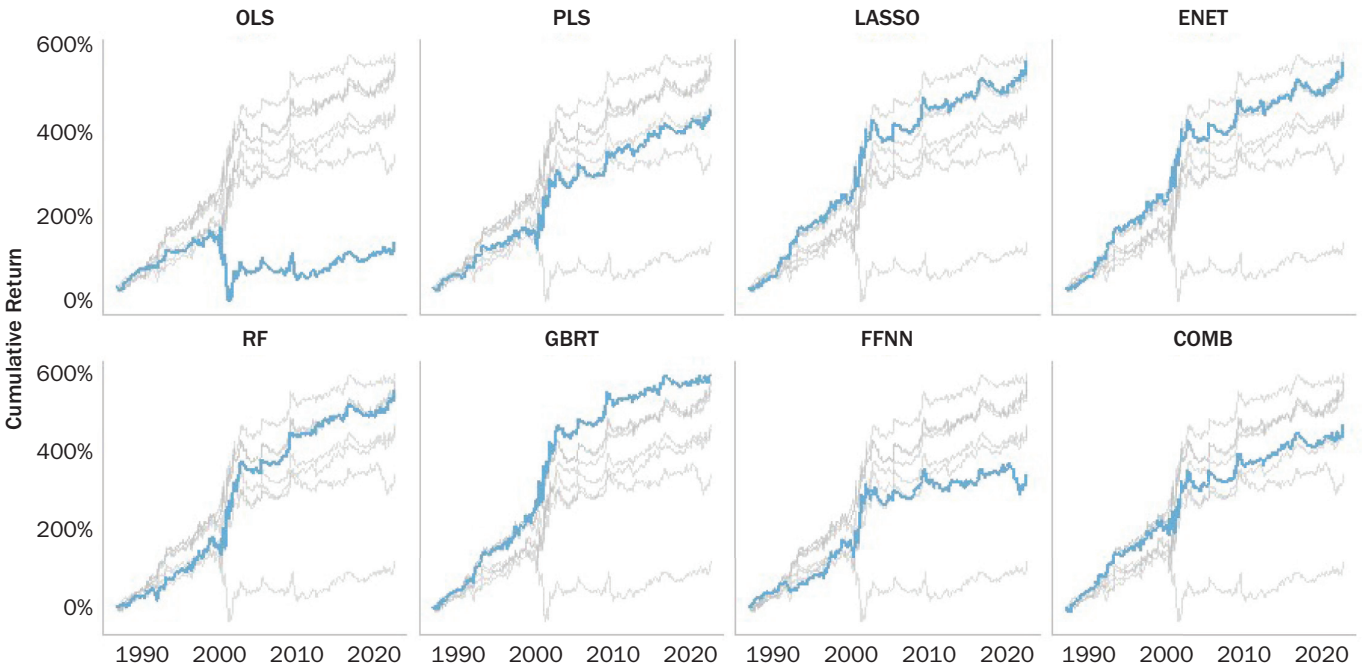
Exhibit 6 plots the cumulative returns on the spread factor portfolios through time. Except for OLS, all models exhibit relatively steady returns over the years, without any major drawdowns. Nonetheless, the profitability in the 1990s and early 2000s appears visually stronger than in more recent years. The decline in return predictability is particularly visible for FFNN but generally affects all considered algorithms.

To formalize these investigations further, we examine the returns of the decile portfolios in subperiods. Specifically, we divide our sample in half and examine the performance in each half. The results of this exercise, reported in Exhibit 7, confirm the visible decline in the predictability of factor returns. Although the spread portfolio returns and their alphas remain significant in both subperiods, their magnitude has

[3] Importantly, the evidence from foreign markets is not always consistent, and Jacobs (2016) and Jacobs and Müller (2020) argue that this phenomenon cannot be generalized internationally.

## EXHIBIT 6

### Cumulative Returns on Machine Learning Portfolios of Factors




NOTES: This exhibit reports cumulative returns on long-short portfolios that buy (sell) the decile factors with the highest (lowest) predicted returns according to different machine learning models: OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). The values are cumulated additively and expressed in percentages. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021. The blue lines represent the model indicated in the heading, whereas the grey are the other models provided for comparison.

been significantly lower in recent years. Without OLS, the average return differentials between the top and bottom deciles ranged from 1.24% to 2.05% over the period from January 1987 to June 2004, and the corresponding alphas were similar in size. In the second half of the study period—between July 2004 and December 2021—the average spread portfolio returns, however, shrank to between 0.30% and 0.95%. In summary, although return predictability remains strong and significant, its magnitude has declined considerably over time.

**Alternative break points.** Our basic approach assumes the use of deciles at each step of the testing procedure. That is, first, the anomaly portfolios are constructed from decile portfolios as individual securities, and second, these anomalies are sorted into deciles to identify the cross-sectional patterns in their returns. As recent discussions on nonstandard errors have shown (Menkveld et al. 2024; Soebhag, Van Vliet, and Verwijmeren 2024; Walter, Weber, and Weiss 2024; Coqueret 2023), however, such a seemingly minor methodological choice can have a material impact on the final results. Therefore, to ensure the robustness of our results, we consider two alternative sorting methods: terciles and quintiles. Notably, we apply them at both levels, that is, individual stocks in anomaly portfolio formation and factors in grouping them based on machine learning predictions. The intersection of the three sorting approaches at these two levels yields nine possible combinations. Exhibit 8 shows the results of this exercise.

For the sake of brevity, we limit the presentation to COMB, which encapsulates the other seven individual models. The results convincingly confirm the predictability of factor returns across all possible approaches. In all cases, the spread portfolios

## EXHIBIT 7

### Machine Learning Factors Portfolios in Subperiods

| | OLS | PLS | LASSO | ENET | RF | GBRT | FFNN | COMB |
|---|---|---|---|---|---|---|---|---|
| **Panel A: January 1987 to June 2004** | | | | | | | | |
| Low | 0.13 | –0.55 | –0.66 | –0.66 | –0.87 | –0.99 | –0.40 | –0.61 |
| 2 | –0.06 | –0.38 | –0.54 | –0.54 | –0.38 | –0.63 | –0.11 | –0.41 |
| 3 | 0.18 | –0.22 | –0.38 | –0.38 | –0.29 | –0.39 | –0.14 | –0.22 |
| 4 | 0.22 | –0.21 | –0.37 | –0.37 | –0.04 | –0.14 | –0.18 | 0.02 |
| 5 | 0.01 | 0.09 | –0.11 | –0.11 | 0.08 | 0.03 | –0.17 | –0.08 |
| 6 | –0.04 | 0.12 | 0.27 | 0.28 | 0.07 | 0.04 | 0.12 | 0.14 |
| 7 | –0.20 | 0.06 | 0.22 | 0.21 | 0.20 | 0.47 | 0.13 | 0.01 |
| 8 | 0.03 | 0.45 | 0.38 | 0.38 | 0.43 | 0.34 | 0.11 | 0.23 |
| 9 | 0.11 | 0.57 | 0.61 | 0.62 | 0.65 | 0.73 | 0.36 | 0.70 |
| High | 0.32 | 0.69 | 1.09 | 1.09 | 0.75 | 1.06 | 0.86 | 0.81 |
| High-Low | 0.19 | 1.24 | 1.75 | 1.75 | 1.62 | 2.05 | 1.26 | 1.43 |
| | *(0.31)* | *(2.40)* | *(3.32)* | *(3.32)* | *(2.82)* | *(4.17)* | *(2.50)* | *(3.07)* |
| $\alpha_{F6}$ | 0.33 | 1.37 | 1.52 | 1.52 | 1.34 | 1.89 | 1.02 | 1.33 |
| | *(0.55)* | *(2.70)* | *(3.11)* | *(3.11)* | *(2.65)* | *(3.59)* | *(2.80)* | *(2.51)* |
| **Panel B: July 2004 to December 2021** | | | | | | | | |
| Low | –0.09 | –0.41 | –0.48 | –0.46 | –0.31 | –0.17 | 0.01 | –0.23 |
| 2 | –0.10 | –0.11 | –0.07 | –0.09 | –0.08 | –0.13 | –0.14 | –0.09 |
| 3 | –0.10 | –0.18 | –0.06 | –0.06 | 0.01 | –0.11 | –0.14 | –0.25 |
| 4 | 0.03 | 0.01 | 0.01 | 0.01 | 0.10 | –0.07 | 0.18 | –0.08 |
| 5 | 0.01 | 0.08 | 0.03 | 0.03 | –0.03 | –0.05 | 0.00 | 0.10 |
| 6 | 0.13 | 0.18 | 0.04 | 0.05 | 0.09 | 0.02 | 0.26 | 0.07 |
| 7 | 0.20 | 0.39 | 0.28 | 0.28 | –0.01 | 0.28 | 0.01 | 0.27 |
| 8 | 0.43 | 0.23 | 0.31 | 0.30 | 0.27 | 0.18 | 0.28 | 0.33 |
| 9 | 0.23 | 0.32 | 0.46 | 0.45 | 0.26 | 0.44 | 0.23 | 0.33 |
| High | 0.26 | 0.47 | 0.45 | 0.45 | 0.64 | 0.56 | 0.32 | 0.50 |
| High-Low | 0.35 | 0.88 | 0.94 | 0.91 | 0.95 | 0.73 | 0.30 | 0.73 |
| | *(0.90)* | *(2.32)* | *(2.13)* | *(2.07)* | *(2.71)* | *(1.91)* | *(0.72)* | *(1.87)* |
| $\alpha_{F6}$ | 0.26 | 1.02 | 1.14 | 1.13 | 1.08 | 0.87 | 0.32 | 0.83 |
| | *(0.83)* | *(3.24)* | *(3.29)* | *(3.25)* | *(3.23)* | *(2.55)* | *(1.24)* | *(2.63)* |

**NOTES:** This exhibit reports monthly returns on decile portfolios of factor strategies formed on the predictions of machine learning models in subperiods. The considered models include OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). Low (high) indicates the decile of factors with the lowest (highest) predicted return. High-Low is the spread portfolio that assumes a long (short) position in the high (low) deciles. $\alpha_{F6B}$ represents the alpha from the six-factor model of Fama and French (2018). The numbers in parentheses are Newey and West's (1987) adjusted *t*-statistics. All the returns and alphas are expressed in percentages. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021. The results are reported for halves of the full testing period: January 1987 to June 2004 (Panel A) and July 2004 to December 2021 (Panel B).

generate large and significant mean returns and alphas. Their magnitude is typically higher for wider spreads, both at the anomaly and the stock level. This is intuitively understandable, however, because the greater distance between the top and bottom quantiles translates into larger differences in expected returns. In summary, the cross-sectional predictability of factor returns is robust across many specifications.

## WHAT DRIVES THE FACTOR RETURN PREDICTABILITY?

In this section, we focus on the main drivers of return predictability in factor returns. First, we perform a variable importance analysis. Next, we zoom in on the role of one specific phenomenon: factor momentum.

## EXHIBIT 8

### Returns on Machine Learning Factors Portfolios Based on Alternative Break Points

| Anomaly-Level Sorts | Deciles | | | Quintiles | | | Terciles | | |
|---|---|---|---|---|---|---|---|---|---|
| Stock-Level Sorts | Deciles | Quintiles | Terciles | Deciles | Quintiles | Terciles | Deciles | Quintiles | Terciles |
| $R$ | 1.08 | 0.90 | 0.58 | 0.96 | 0.74 | 0.51 | 0.75 | 0.58 | 0.39 |
| | *(3.51)* | *(3.72)* | *(3.07)* | *(3.70)* | *(3.53)* | *(3.23)* | *(3.66)* | *(3.56)* | *(3.23)* |
| $\alpha_{F6}$ | 1.04 | 0.92 | 0.62 | 0.93 | 0.76 | 0.54 | 0.73 | 0.58 | 0.43 |
| | *(3.37)* | *(3.61)* | *(2.86)* | *(3.66)* | *(3.47)* | *(3.01)* | *(3.51)* | *(3.31)* | *(2.92)* |
| $\alpha_{B}$ | 1.20 | 0.98 | 0.63 | 1.06 | 0.82 | 0.55 | 0.83 | 0.64 | 0.43 |
| | *(3.29)* | *(3.47)* | *(2.85)* | *(3.41)* | *(3.36)* | *(3.00)* | *(3.31)* | *(3.29)* | *(2.86)* |

**NOTES:** This exhibit reports monthly returns on long–short portfolios that buy (sell) factors with the highest (lowest) predicted returns according to the forecast combination model (COMB), which aggregates seven individual machine learning models: OLS, PLS, LASSO, ENET, random forests (RF), GBRT, and FFNN. $R$ is the mean monthly return. $\alpha_{F6}$ and $\alpha_{B}$ are alphas from the six-factor model of Fama and French (2018) and the factor benchmark model, respectively. The results are reported for different levels of break points used to form both anomaly portfolios of stocks as well as machine learning portfolios of factor strategies; as indicated in the top rows, at both levels we consider sorts into deciles, quintiles, and terciles. All the returns and alphas are expressed in percentages. The numbers in parentheses are Newey and West's (1987) adjusted $t$-statistics. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

### Variable Importance

To better understand the sources of cross-sectional variation in factor returns, we now turn to the analysis of characteristic importance. We aim to identify which features contribute the most to the return predictability. To capture this, we compute variable importance (VI) as in Kelly, Pruitt, and Su (2019). Precisely, for each factor characteristic, we calculate the decrease in $R^2_{OOS}$ resulting from setting a given variable to zero while keeping all else fixed. By doing this, we are able to pinpoint the critical features that influence the cross-sectional variation of factor returns accounting for all 242 signals within the system. To facilitate better presentation, we rescale VI so that their sum equals one.

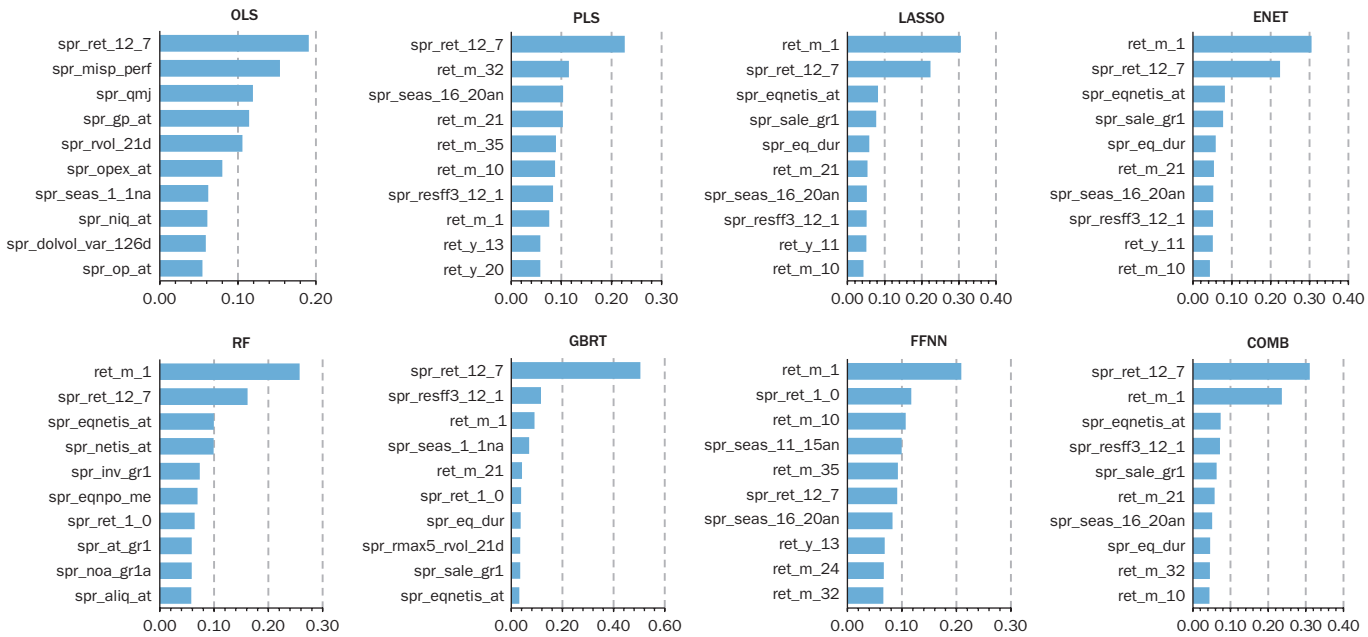
Exhibit 9 illustrates the top 10 features per model. Notably, most of the models agree on the selection of the critical features, pointing out largely consistent variables. The top characteristics are typically either related to past factor returns, such as *ret_m_1*, *ret_m_21*, or *ret_m_10*, or to stock momentum. The latter group refers to factor spreads based on momentum-related variables, with the intermediate momentum (*spr_ret_12_7*) playing the most important role. The influence of factor momentum, in turn, emphasizes particularly the most recent month (*ret_m_1*), which was also deemed crucial by Gupta and Kelly (2019). Among the nonmarket variables, the prime role is played by factor spreads associated with equity issuance (*spr_eqnetis_at*) and sales growth (*spr_eq_g1*), though, admittedly, their importance is noticeably lower than the momentum features.

Interestingly, the distribution of variable importance is far from democratic but rather concentrated in a handful of key variables. Exhibit 10 shows the aggregate importance of the top 3, 5, and 10 features, calculated as in Bali et al. (2020). Depending on the model, the aggregate importance of just three top features ranges from 6% (FFNN) to 42% (GBRT). For the well-performing penalized regressions, the top three features account for as much as 34% of the return predictability. Increasing the number of variables to 10 increases the overall importance to 55%. Furthermore, for the GBRT—the best-performing model in many previous tests—the aggregate importance of the top 10 variables is 59%.

In summary, the variable importance analysis suggests that a large portion of return predictability comes from just a few return predictors. Moreover, these few key

## EXHIBIT 9

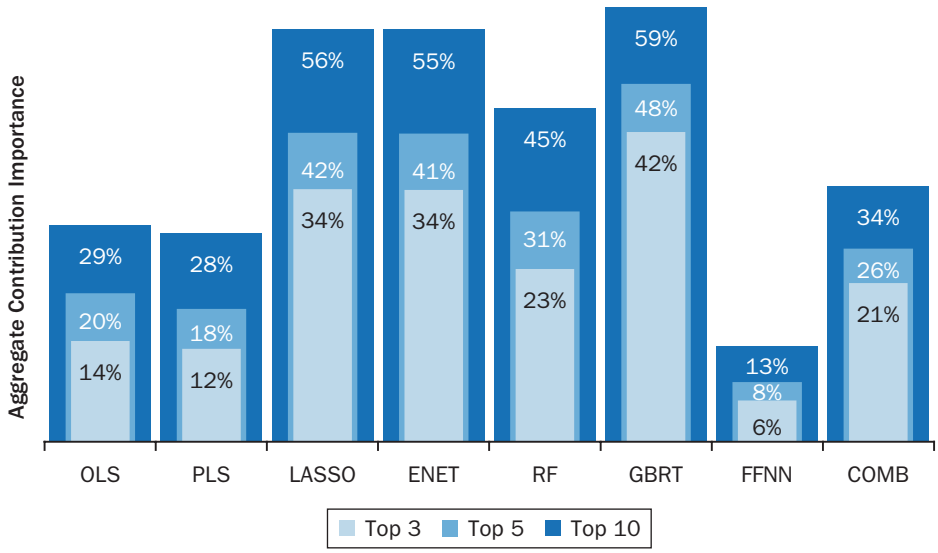
### Variable Importance




**NOTES:** This exhibit presents the importance of the top 10 features in each of the machine learning models considered: OLS, PLS, LASSO, ENET, GBRT, random forests (RF), FFNN, and forecast combination (COMB). The VI is calculated as the reduction in $R^2_{OOS}$ due to setting all values of a given feature in the training sample to zero. The reported VI is averaged across all the training samples and rescaled to sum to 1 across all characteristics. The sample contains 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

## EXHIBIT 10

### Aggregate Importance of Top Variables




**NOTES:** This exhibit illustrates the aggregate contribution of the top 3, 5, and 10 predictors for the eight machine learning models considered: OLS, PLS, LASSO, ENET, GBRT, random forests (RF), FFNN, and forecast combination (COMB). The VI is calculated as the reduction in $R^2_{OOS}$ due to setting all values of a given feature in the training sample to zero. The reported VI is averaged across all the training samples and rescaled to sum to 1 across all characteristics. The aggregate importance is obtained as in Bali et al. (2020). The sample contains 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

features typically include some form of momentum—or, more specifically, momentum in anomaly returns. This highlights the special role of factor momentum, which tops in four of the eight VI rankings. This begs the question: What is the role of factor momentum in the overall predictability of factor returns?

### The Role of Factor Momentum

The VI analysis highlighted the special role of factor momentum in the predictability of factor returns. To investigate this further, we examine how much of the cross-sectional variation can be attributed to factor momentum. To address this issue, we conduct spanning tests. Specifically, we regress the returns of machine learning portfolios against the returns of factor momentum strategies.

To this end, we form a simple factor momentum strategy in the spirit of Ehsani and Linnainmaa (2022); Arnott, Kalesnik, and Linnainmaa (2023); and Leippold and Yang (2021). For each factor $i$ at month $t$, we calculate the factor momentum signal $FM_{i,t}$ as

$$FM_{i,t} = \bar{R}_{i,t-12:t-1} - \bar{R}_{i,1:t-1} \tag{6}$$

where $\bar{R}_{i,t-12:t-1}$ is the mean returnover the last 12 months, and $\bar{R}_{i,1:t-1}$ is the prevailing (or historic) mean based on all data available up to month $t-1$. The 12-month sorting period originates from the standard approach in Ehsani and Linnainmaa (2022). The adjustment with the long-term mean, in turn, builds on the philosophy of Leippold and Yang (2021), aiming at disentangling the unconditional expected return (buy-and-hold component) from the actual factor momentum (market timing component). Notably, although we employ this correction to assure the prudence of our approach, it has no material impact on the overall findings—the results without the adjustment are fully consistent.

With the factor momentum signals at hand, we form the cross-sectional momentum factor portfolio $FMOM_t$. To be specific, each month, we sort factors into terciles based on $FM_{i,t}$. Next, we construct a spread strategy that is long in the factors with the highest $FM_{i,t}$ and short in those with the lowest signal. Notably, we use cross-sectional factor momentum rather than its time-series component to possibly closely align with the testing strategies—the time-series momentum exhibits time-varying allocation to its long and short legs, while the tested machine learning strategies incorporate no such timing component.

Last, we regress the returns on the machine learning portfolios against the factor momentum strategies:

$$R_t = \alpha + \beta_{FMOM} FMOM_t + \varepsilon_t \tag{7}$$

where $R_t$ is the return on the tested portfolio of factors in month $t$, $FMOM_t$ is the factor momentum portfolio return, and $\varepsilon_t$ is the error term. $\alpha$ and $\beta_{FMOM}$ are the estimated regression parameters. We aim to see to what extent the exposure to the factor momentum ($\beta_{FMOM}$) accounts for machine learning profits.

Exhibit 11 reports the results of the regressions, with Panel A reporting $\beta_{FMOM}$ and Panel B focusing on $\alpha$. Starting with Panel A, the machine learning portfolios display the evident exposure to factor momentum portfolios. The slope coefficients for bottom deciles are typically negative and then gradually increase along with the return forecast to become visibly positive for the top decile. In consequence, the spread portfolios also exhibit a significantly positive exposure, with $\beta_{FMOM}$ ranging from 0.29 (OLS) to 0.81 (LASSO, ENET). In other words, the machine learning strategy returns visibly comove with factor momentum.

How much profitability is captured by FMOM exposure? Panel B sheds light on this question. In a nutshell, controlling for FMOM explains virtually all of the abnormal returns of the machine learning portfolios. Most decile portfolios no longer exhibit significant alpha, including all top and all bottom deciles. More importantly, applying Model 7 also renders all alphas of the spread portfolios insignificant. Put differently, once we control for the factor momentum, no strategy based on machine learning forecasts generates reliable abnormal returns.

The evidence in Exhibit 11, combined with the earlier results of variable importance, suggests that factor momentum is the critical driver of cross-sectional variation in factor returns. Although the literature has documented many signals, factor momentum subsumes their machine learning aggregation and emerges as the key determinant of return predictability in factor returns.

To further validate our conclusions from Exhibit 11, we run three simple robustness checks. First, we split our sample into two equal subperiods, analogous to Exhibit 7. Second, we consider alternative break points at the security level to form anomaly portfolios: quintiles and terciles. Third, we consider the same groupings—quintiles and terciles—at the level of machine learning portfolios applied to factors. Exhibit 12 summarizes the results of these exercises. For brevity, we report only the results for the spread portfolios.

## EXHIBIT 11

### Machine Learning Predictions vs. Factor Momentum

| | OLS | PLS | LASSO | ENET | RF | GBRT | FFNN | COMB |
|---|---|---|---|---|---|---|---|---|
| **Panel A: Factor Momentum Exposure ($\beta_{FMOM}$)** | | | | | | | | |
| Low | –0.17 | –0.27 | –0.39 | –0.39 | –0.39 | –0.46 | –0.11 | –0.36 |
| | *(–1.42)* | *(–2.15)* | *(–3.13)* | *(–3.13)* | *(–2.92)* | *(–3.57)* | *(–0.92)* | *(–2.83)* |
| 2 | –0.14 | –0.20 | –0.27 | –0.26 | –0.18 | –0.28 | –0.02 | –0.21 |
| | *(–1.79)* | *(–2.12)* | *(–3.06)* | *(–3.03)* | *(–1.88)* | *(–3.28)* | *(–0.25)* | *(–2.41)* |
| 3 | –0.04 | –0.12 | –0.22 | –0.22 | –0.15 | –0.19 | –0.14 | –0.07 |
| | *(–0.85)* | *(–1.66)* | *(–3.37)* | *(–3.37)* | *(–1.90)* | *(–3.00)* | *(–1.91)* | *(–0.95)* |
| 4 | 0.05 | –0.12 | –0.22 | –0.23 | 0.08 | –0.08 | –0.16 | –0.11 |
| | *(1.13)* | *(–1.90)* | *(–4.22)* | *(–4.29)* | *(1.10)* | *(–1.60)* | *(–2.71)* | *(–2.33)* |
| 5 | 0.00 | 0.05 | –0.03 | –0.03 | –0.02 | –0.11 | –0.14 | –0.08 |
| | *(–0.05)* | *(1.04)* | *(–0.63)* | *(–0.64)* | *(–0.48)* | *(–2.46)* | *(–2.72)* | *(–1.58)* |
| 6 | –0.15 | 0.00 | 0.09 | 0.10 | –0.13 | –0.01 | –0.04 | –0.02 |
| | *(–3.02)* | *(–0.12)* | *(2.22)* | *(2.36)* | *(–2.29)* | *(–0.39)* | *(–0.75)* | *(–0.43)* |
| 7 | –0.04 | –0.04 | 0.01 | 0.01 | –0.03 | 0.21 | 0.05 | 0.01 |
| | *(–0.71)* | *(–0.90)* | *(0.27)* | *(0.23)* | *(–0.63)* | *(4.96)* | *(0.82)* | *(0.30)* |
| 8 | 0.07 | 0.11 | 0.13 | 0.13 | 0.09 | 0.08 | 0.09 | 0.05 |
| | *(0.98)* | *(1.87)* | *(2.27)* | *(2.19)* | *(1.47)* | *(1.21)* | *(1.67)* | *(0.75)* |
| 9 | 0.07 | 0.20 | 0.18 | 0.19 | 0.19 | 0.30 | –0.05 | 0.21 |
| | *(0.89)* | *(2.75)* | *(2.20)* | *(2.22)* | *(2.28)* | *(3.99)* | *(–0.62)* | *(2.80)* |
| High | 0.12 | 0.15 | 0.42 | 0.42 | 0.29 | 0.28 | 0.26 | 0.30 |
| | *(1.07)* | *(1.56)* | *(3.89)* | *(3.90)* | *(2.53)* | *(2.67)* | *(2.53)* | *(2.92)* |
| High-Low | 0.29 | 0.42 | 0.81 | 0.81 | 0.68 | 0.74 | 0.36 | 0.66 |
| | *(1.33)* | *(2.01)* | *(3.82)* | *(3.83)* | *(3.01)* | *(3.42)* | *(1.96)* | *(3.13)* |

*(continued)*

## EXHIBIT 11 *(continued)*

### Machine Learning Predictions vs. Factor Momentum

| | OLS | PLS | LASSO | ENET | RF | GBRT | FFNN | COMB |
|---|---|---|---|---|---|---|---|---|
| **Panel B: Average Unexplained Return (α)** | | | | | | | | |
| Low | 0.52 | 0.30 | 0.58 | 0.59 | 0.55 | 0.77 | 0.12 | 0.62 |
| | *(1.27)* | *(0.71)* | *(1.36)* | *(1.38)* | *(1.21)* | *(1.75)* | *(0.31)* | *(1.45)* |
| 2 | 0.34 | 0.35 | 0.48 | 0.46 | 0.30 | 0.44 | –0.06 | 0.37 |
| | *(1.25)* | *(1.08)* | *(1.60)* | *(1.56)* | *(0.92)* | *(1.51)* | *(–0.21)* | *(1.23)* |
| 3 | 0.17 | 0.14 | 0.43 | 0.42 | 0.30 | 0.30 | 0.26 | –0.04 |
| | *(0.94)* | *(0.59)* | *(1.92)* | *(1.89)* | *(1.11)* | *(1.41)* | *(1.05)* | *(–0.16)* |
| 4 | –0.03 | 0.24 | 0.47 | 0.48 | –0.19 | 0.14 | 0.48 | 0.31 |
| | *(–0.20)* | *(1.17)* | *(2.62)* | *(2.68)* | *(–0.82)* | *(0.78)* | *(2.34)* | *(1.85)* |
| 5 | 0.02 | –0.08 | 0.04 | 0.05 | 0.10 | 0.32 | 0.33 | 0.24 |
| | *(0.11)* | *(–0.44)* | *(0.29)* | *(0.30)* | *(0.58)* | *(2.07)* | *(1.86)* | *(1.42)* |
| 6 | 0.48 | 0.16 | –0.11 | –0.12 | 0.47 | 0.07 | 0.29 | 0.16 |
| | *(2.87)* | *(1.19)* | *(–0.80)* | *(–0.84)* | *(2.37)* | *(0.58)* | *(1.79)* | *(1.05)* |
| 7 | 0.11 | 0.35 | 0.21 | 0.22 | 0.19 | –0.23 | –0.06 | 0.10 |
| | *(0.61)* | *(2.10)* | *(1.38)* | *(1.40)* | *(1.08)* | *(–1.64)* | *(–0.32)* | *(0.61)* |
| 8 | 0.04 | 0.02 | –0.05 | –0.03 | 0.08 | 0.03 | –0.08 | 0.14 |
| | *(0.16)* | *(0.11)* | *(–0.23)* | *(–0.17)* | *(0.37)* | *(0.13)* | *(–0.40)* | *(0.63)* |
| 9 | –0.04 | –0.15 | 0.00 | –0.01 | –0.09 | –0.30 | 0.44 | –0.10 |
| | *(–0.14)* | *(–0.59)* | *(0.00)* | *(–0.03)* | *(–0.31)* | *(–1.17)* | *(1.63)* | *(–0.40)* |
| High | –0.06 | 0.14 | –0.44 | –0.45 | –0.15 | –0.01 | –0.16 | –0.23 |
| | *(–0.16)* | *(0.44)* | *(–1.22)* | *(–1.23)* | *(–0.39)* | *(–0.04)* | *(–0.47)* | *(–0.65)* |
| High-Low | –0.58 | –0.16 | –1.02 | –1.04 | –0.70 | –0.78 | –0.29 | –0.85 |
| | *(–0.78)* | *(–0.23)* | *(–1.42)* | *(–1.44)* | *(–0.92)* | *(–1.06)* | *(–0.45)* | *(–1.19)* |

**NOTES:** This exhibit reports the results of the spanning regressions of the returns on machine learning factor strategies on the returns on the factor momentum portfolios:

$$R_t = \alpha + \beta_{FMOM} FMOM_t + \varepsilon_t$$

where $R_t$ is the return on the tested portfolio of factors in month $t$, $FMOM_t$ is the return on the factor momentum portfolio, and $\varepsilon_t$ is the error term. $\alpha$ and $\beta_{FMOM}$ are the estimated regression parameters, with Panel A presenting the measures of factor momentum exposure ($\beta_{FMOM}$) and Panel B showing the average monthly unexplained return ($\alpha$). The portfolios of factors are sorted based on predictions from eight machine learning models: OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). Low (high) indicates the decile of factors with the lowest (highest) predicted return. High-Low is the spread portfolio that assumes a long (short) position in the high (low) deciles. The numbers in parentheses are Newey and West's (1987) adjusted *t*-statistics. The alphas in Panel B are expressed in percentages. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021.

Our results remain robust to all these alternative considerations. In other words, in all cases considered, no alpha is positive and significant. The abnormal returns are mostly insignificant in all specifications, indicating that the factor momentum fully subsumes the machine learning returns. The few exceptions are the LASSO, ENET, and FFNN portfolios based on quintile anomaly portfolios, for which the alphas are even significantly negative. To conclude, factor momentum takes center stage as a critical driver of predictability, effectively capturing the cross-sectional variation in factor returns.

## EXHIBIT 12

### Machine Learning Predictions vs. Factor Momentum: Robustness Checks

| | OLS | PLS | LASSO | ENET | RF | GBRT | FFNN | COMB |
|---|---|---|---|---|---|---|---|---|
| **Panel A: Subperiods** | | | | | | | | |
| January 1987–June 2004 | –0.77 | –0.08 | –0.79 | –0.79 | –0.42 | –0.29 | –0.20 | –0.72 |
| | *(–0.65)* | *(–0.08)* | *(–0.74)* | *(–0.75)* | *(–0.35)* | *(–0.25)* | *(–0.22)* | *(–0.65)* |
| July 2004–December 2021 | –0.51 | –0.16 | –1.09 | –1.11 | –1.12 | –0.76 | 0.89 | –0.67 |
| | *(–0.52)* | *(–0.16)* | *(–0.94)* | *(–0.95)* | *(–1.05)* | *(–0.71)* | *(0.85)* | *(–0.63)* |
| **Panel B: Alternative Stock-Level Break Points** | | | | | | | | |
| Quintiles | –0.62 | –0.56 | –1.28 | –1.28 | –0.81 | –1.06 | –0.82 | –0.97 |
| | *(–1.06)* | *(–1.01)* | *(–2.23)* | *(–2.23)* | *(–1.25)* | *(–1.86)* | *(–2.18)* | *(–1.77)* |
| Terciles | –0.60 | –0.28 | –0.58 | –0.58 | –0.60 | –0.17 | –0.02 | –0.60 |
| | *(–1.34)* | *(–0.61)* | *(–1.21)* | *(–1.21)* | *(–1.21)* | *(–0.35)* | *(–0.09)* | *(–1.40)* |
| **Panel C: Alternative Anomaly-Level Break Points** | | | | | | | | |
| Quintiles | –0.51 | –0.20 | –0.78 | –0.76 | –0.57 | –0.80 | 0.13 | –0.66 |
| | *(–0.83)* | *(–0.33)* | *(–1.26)* | *(–1.22)* | *(–0.88)* | *(–1.32)* | *(0.23)* | *(–1.09)* |
| Terciles | –0.37 | –0.17 | –0.69 | –0.69 | –0.38 | –0.59 | –0.08 | –0.42 |
| | *(–0.76)* | *(–0.35)* | *(–1.40)* | *(–1.40)* | *(–0.73)* | *(–1.23)* | *(–0.17)* | *(–0.86)* |

**NOTES:** This exhibit reports the robustness checks for the unexplained returns, or alphas ($\alpha$), from the spanning regressions of the returns on long–short machine learning factor strategies on the returns on the factor momentum portfolios:

$$R_t = \alpha + \beta_{FMOM} FMOM_t + \varepsilon_t$$

where $R_t$ is the return on the tested portfolio of factors in month $t$, $FMOM_t$ is the return on the factor momentum portfolio, $\alpha$ and $\beta_{FMOM}$ are the estimated regression parameters, and $\varepsilon_t$ is the error term. The portfolios of factors are sorted based on predictions from eight machine learning models: OLS, PLS, LASSO, ENET, random forests (RF), GBRT, FFNN, and forecast combination (COMB). The numbers in parentheses are Newey and West's (1987) adjusted $t$-statistics. All alphas are expressed in percentages. The sample comprises 153 factors from Jensen, Kelly, and Pedersen (2023); the study period begins in January 1972, and the testing period runs from January 1987 to December 2021. The robustness checks consider different departures from the baseline methodology, that is, tests in subperiods (Panel A) and alternative break points to form the factor portfolios of stocks (Panel B) and machine learning portfolios of factors (Panel C).

## CONCLUSION

In this article, we examine the cross-sectional predictability of factor returns. To this end we compute a comprehensive set of long–short anomaly portfolios using more than five decades of US stock market returns. Using this dataset, we compute 242 factor characteristics that reflect their past returns, risks, and spreads. Finally, we apply machine learning techniques to extract information about future factor returns.

Our results reveal substantial cross-sectional return predictability in the equity factor universe. The $R^2_{OOS}$ for even simple models, such as penalized regressions, are at least on par with the return predictability of individual stocks. Accounting for nonlinearities and interactions, however, adds little value.

In particular, return predictability can be forged into effective factor selection strategies based on portfolio sorting. The top decile of factors outperforms the worst decile by between 0.78% and 1.39% per month, depending on the forecasting model used. This outperformance is remarkably robust, surviving across different factor models and alternative construction methods. The magnitude of the return predictability, however, while still significant, diminishes remarkably over time.

Finally, our research also uncovers the sources of return predictability. VI analysis identifies short-term factor momentum and stock price momentum as the key drivers of predictive performance. Moreover, spanning tests suggest that factor momentum

plays a key role in shaping the cross section of factor returns. When controlled for, it explains all of the abnormal returns of the machine learning long–short factor portfolios. In other words, although the return predictability of factor returns is strong, it is mainly driven by the factor momentum effect.

Our findings have clear practical implications for portfolio managers with a quantitative mandate. Factor characteristics—including factor momentum in particular—contain valuable information about their future returns, allowing one to separate the wheat from the chaff. As a result, one can potentially pick the future winners and avoid the losers in the factor space.

Future research on the topics discussed in this article can be pursued in at least two directions. First, it would be valuable to extend the analyses to international markets. This would shed light on the country and institutional characteristics that shape factor return predictability. Second, although our article focuses on equities, other asset classes—such as corporate bonds, commodities, or currencies—also exhibit multidimensional return structures. The question of their factor return predictability remains open.

# APPENDIX

## STOCK CHARACTERISTICS USED IN THIS ARTICLE

The appendix presents the 153 stock characteristics used to form factor portfolios examined in this article. Detailed calculation methods and source literature are available from Jensen, Kelly, and Pedersen (2023). The exhibit spans two pages.

| Symbol | Characteristic | Symbol | Characteristic |
|---|---|---|---|
| age | Firm age | ni_inc8q | # quarters with earnings increases |
| aliq_at | Liquidity of book assets | ni_ivol | Earnings volatility |
| aliq_mat | Liquidity of market assets | ni_me | Earnings-to-price |
| ami_126d | Amihud Measure | niq_at | Quarterly return on assets |
| at_be | Book leverage | niq_at_chg1 | Change in quarterly return on assets |
| at_gr1 | Asset growth | niq_be | Quarterly return on equity |
| at_me | Assets-to-market | niq_be_chg1 | Change in quarterly return on equity |
| at_turnover | Capital turnover | niq_su | Standardized earnings surprise |
| be_gr1a | Change in common equity | nncoa_gr1a | Change in net noncurrent oper. assets |
| be_me | Book-to-market equity | noa_at | Net operating assets |
| beta_60m | Market beta | noa_gr1a | Change in net operating assets |
| beta_dimson_21d | Dimson beta | o_score | Ohlson O-score |
| betabab_1260d | Frazzini-Pedersen market beta | oaccruals_at | Operating accruals |
| betadown_252d | Downside beta | oaccruals_ni | Percentage operating accruals |
| bev_mev | Book-to-market enterprise value | ocf_at | Operating cash flow to assets |
| bidaskhl_21d | The high-low bid–ask spread | ocf_at_chg1 | Change in operating cash flow to assets |
| capex_abn | Abnormal corporate investment | ocf_me | Operating cash flow-to-market |
| capx_gr1 | CAPEX growth (1 year) | ocfq_saleq_std | Cash flow volatility |
| capx_gr2 | CAPEX growth (2 years) | op_at | Operating profits-to-book assets |
| capx_gr3 | CAPEX growth (3 years) | op_atl1 | Operating profits-to-lagged book assets |
| cash_at | Cash-to-assets | ope_be | Operating profits-to-book equity |
| chcsho_12m | Net stock issues | ope_bel1 | Operating profits-to-lagged book equity |
| coa_gr1a | Change in current operating assets | opex_at | Operating leverage |
| col_gr1a | Change in current operating liabilities | pi_nix | Taxable income-to-book income |
| cop_at | Cash-based oper. profits-to-book assets | ppeinv_gr1a | Change PPE and Inventory |
| cop_atl1 | Cash oper. prof.-to-lagged book assets | prc | Price per share |

*(continued)*

| Symbol | Characteristic | Symbol | Characteristic |
|---|---|---|---|
| corr_1260d | Market correlation | prc_highpr_252d | Current price to high price over last year |
| coskew_21d | Coskewness | qmj | Quality minus junk: composite |
| cowc_gr1a | Change in current operating working capital | qmj_growth | Quality minus junk: growth |
| dbnetis_at | Net debt issuance | qmj_prof | Quality minus junk: profitability |
| debt_gr3 | Growth in book debt (3 years) | qmj_safety | Quality minus junk: safety |
| debt_me | Debt-to-market | rd_me | R&D-to-market |
| dgp_dsale | Change gross margin minus change sales | rd_sale | R&D-to-sales |
| div12m_me | Dividend yield | rd5_at | R&D capital-to-book assets |
| dolvol_126d | Dollar trading volume | resff3_12_1 | Residual momentum *t*–12 to *t*–1 |
| dolvol_var_126d | Coef. of variation for $ trading volume | resff3_6_1 | Residual momentum *t*–6 to *t*–1 |
| dsale_dinv | Change sales minus change inventory | ret_1_0 | Short-term reversal |
| dsale_drec | Change sales minus change receivables | ret_12_1 | Price momentum *t*–12 to *t*–1 |
| dsale_dsga | Change sales minus change SG&A | ret_12_7 | Price momentum *t*–12 to *t*–7 |
| earnings_variab | Earnings variability | ret_3_1 | Price momentum *t*–3 to *t*–1 |
| ebit_bev | Return on net operating assets | ret_6_1 | Price momentum *t*–6 to *t*–1 |
| ebit_sale | Profit margin | ret_60_12 | Long-term reversal |
| ebitda_mev | EBITDA-to-market enterprise value | ret_9_1 | Price momentum *t*–9 to *t*–1 |
| emp_gr1 | Hiring rate | rmax1_21d | Maximum daily return |
| eq_dur | Equity duration | rmax5_21d | Highest 5 days of return |
| eqnetis_at | Net equity issuance | rmax5_rvol_21d | Highest 5 days of return scaled by volatility |
| eqnpo_12m | Equity net payout | rskew_21d | Total skewness |
| eqnpo_me | Net payout yield | rvol_21d | Return volatility |
| eqpo_me | Payout yield | sale_bev | Assets turnover |
| f_score | Pitroski F-score | sale_emp_gr1 | Labor force efficiency |
| fcf_me | Free cash flow-to-price | sale_gr1 | Sales growth (1 year) |
| fnl_gr1a | Change in financial liabilities | sale_gr3 | Sales growth (3 years) |
| gp_at | Gross profits-to-assets | sale_me | Sales-to-market |
| gp_atl1 | Gross profits-to-lagged assets | saleq_gr1 | Sales growth (1 quarter) |
| inv_gr1 | Inventory growth | saleq_su | Standardized revenue surprise |
| inv_gr1a | Inventory change | seas_1_1an | Year 1-lagged return, annual |
| iskew_capm_21d | Idiosyncratic skewness (CAPM) | seas_1_1na | Year 1-lagged return, nonannual |
| iskew_ff3_21d | Id. skew. (Fama–French 3-factor model) | seas_11_15an | Years 11–15 lagged returns, annual |
| iskew_hxz4_21d | Idiosyncratic skewness (q-factor model) | seas_11_15na | Years 11–15 lagged returns, nonannual |
| ival_me | Intrinsic value-to-market | seas_16_20an | Years 16–20 lagged returns, annual |
| ivol_capm_21d | Idiosyncratic volatility (CAPM, (21 days) | seas_16_20na | Years 16–20 lagged returns, nonannual |
| ivol_capm_252d | Idiosyncratic volatility (CAPM, 252 days) | seas_2_5an | Years 2–5 lagged returns, annual |
| ivol_ff3_21d | Id. vol. (Fama–French 3-factor model) | seas_2_5na | Years 2–5 lagged returns, nonannual |
| ivol_hxz4_21d | Idiosyncratic volatility (q-factor model) | seas_6_10an | Years 6–10 lagged returns, annual |
| kz_index | Kaplan–Zingales index | seas_6_10na | Years 6–10 lagged returns, nonannual |
| lnoa_gr1a | Change in long-term net operating assets | sti_gr1a | Change in short-term investments |
| lti_gr1a | Change in long-term investments | taccruals_at | Total accruals |
| market_equity | Market equity | taccruals_ni | Percent total accruals |
| mispricing_mgmt | Mispricing factor: management | tangibility | Asset tangibility |
| mispricing_perf | Mispricing factor: performance | tax_gr1a | Tax expense surprise |
| ncoa_gr1a | Change in noncurrent operating assets | turnover_126d | Share turnover |
| ncol_gr1a | Change in noncurrent operating liabilities | turnover_var_126d | Coef. of variation for share turnover |
| netdebt_me | Net debt-to-price | z_score | Altman Z-score |
| netis_at | Net total issuance | zero_trades_126d | No. zero trades (6 months) |
| nfna_gr1a | Change in net financial assets | zero_trades_21d | No. zero trades (1 month) |
| ni_ar1 | Earnings persistence | zero_trades_252d | No. zero trades (12 months) |
| ni_be | Return on equity | | |

## ACKNOWLEDGMENT

Adam Zaremba acknowledges the support of the National Science Center of Poland (Grant No. 2022/45/B/HS4/00451).

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